Selected follow-up questions

Aleph Platinum · Tutor-controlled question sheets

Do not distribute this entire book before testing. Release only the one or two question pages selected for a session. The IDs are for recording; they are not hints about the method. No answers appear here.

A1-4

Solve over the reals: $\sqrt{(x + 6)} = x$. Explain how you reject any extra candidate.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A1-R

Solve x(x + 2) = 0. A student divides by x and reports only −2. What is missing?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A2-4

P is monic with integer coefficients and $P(0)=-1$, $P(1)=1$. Can P have a rational root? Justify.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A2-R

For $3x^2 - 5x + 2$, test $x=1$ and use the result to factor the polynomial.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A3-4

If r,s solve $x^2 - 5x + 3 = 0$, find $1/r^3 + 1/s^3$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A3-R

Roots u,v of $x^2 - 6x + 4$ satisfy what value of u^2+v^2?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A4-4

Find the number of real solutions of $x^4 - 3x^3 + 4x^2 - 3x + 1 = 0$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A4-R

If $x+1/x=4$, $x \neq 0$, find x^2+1/x^2.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A5-4

Solve $\log_2(x-1)+\log_2(x+1)=3$ over the reals.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

A5-R

Solve $3^{x-2}=27$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I1-4

Find all real x for which $x^2 \le 2x+3$ and $\frac{x-2}{x+1} > 0$ both hold.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I1-R

Solve $-2x \geq 8$. Explain the direction of the answer.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I2-4

For which real t does $|x-1|+|x-5|=t$ have exactly two real solutions x?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I2-R

Solve $|x-4| \leq 2$ and describe the interval in words.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I3-4

Positive x,y satisfy xy=4. Find the minimum of (x+y)(x+4y), with equality values.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

I3-R

Positive a,b have a+b=8. Find the maximum ab.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

F1-4

For real x, f(x)=(x−1)/(x+1). Determine the domain of f(f(x)) and simplify it.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

F1-R

If $f(x)=x+1$ and $g(x)=x^2$, find $f(g(3))$ and $g(f(3))$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

F2-4

A function $f:R \to R$ satisfies $f(x+y)=f(x)+f(y)+2xy$ for all real x,y, and $f(1)=3$. Determine $f(r)$ for every rational r.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

F2-R

If $f(x+y)=f(x)+f(y)$ on $\mathbb{R}$ and $f(1)=4$, find $f(3)$ and $f(-1)$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

L1-4

Integers a,b satisfy: ab is divisible by 6. Must at least one of a,b be divisible by 6? Give a counterexample or proof, and explain which prime-divisibility fact still holds.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

L1-R

For integers n, is “3 divides n implies 6 divides n” true? Justify.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

L2-4

For real a, when is this true: for every real x there exists real y such that $y^2=x+a$? Negate the statement and decide it for every a.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

L2-R

Negate “Every student solved at least one question.” Keep the student/question domains unchanged.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S1-4

A list of consecutive positive integers starts at 1 and has sum 210. Delete one number and the mean of the remaining numbers becomes 10. Which number was deleted?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S1-R

Find 5+8+11+⋯+32.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S2-4

Positive integers m,n satisfy $1+2+\cdots+2^{n-1}=3(1+4+\cdots+4^{m-1})$. Find all ordered pairs (m,n).

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S2-R

Find 1+2+4+8+16+32.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S3-4

Without assuming the sum-of-squares formula, derive it by summing $(k+1)^3 - k^3$ for $k=1,\ldots,n$. You may use $\Sigma k = n(n+1)/2$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S3-R

Find $1^2+2^2+\cdots+5^2$, and separately find $1+2+\cdots+5$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S4-4

Let $a_n = n(n+1)(n+2)$. Find $a_{n+1} - a_n$ and hence evaluate $\Sigma\, k(k+1)$ for $k=1,\ldots,n$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S4-R

Find (1−1/2)+(1/2−1/3)+(1/3−1/4).

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S5-4

For positive integer n, derive a formula for $\Sigma\, k \cdot 2^k$, k=1,…,n.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

S5-R

Compute $1 \cdot 3 + 2 \cdot 3^2 + 3 \cdot 3^3$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C1-4

How many six-digit positive integers use each of 0,1,2,3,4,5 exactly once and are divisible by 5?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C1-R

A shirt has 4 colour choices and 3 size choices; every combination exists. How many combinations?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C2-4

From 6 women and 5 men, choose a committee of 4 with at least 2 women, then appoint one committee member as chair. How many outcomes?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C2-R

From 5 distinct people choose a pair, then distinguish one as leader. How many outcomes?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C3-4

How many integers from 1 through 200 are divisible by at least one of 2,3,5?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C3-R

Twelve students play chess, 10 play football and 4 play both. How many play at least one?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C4-4

Prove $\Sigma\, k \cdot C(n,k) = n \cdot 2^{n-1}$, $k=0,\ldots,n$, for positive integer n, by counting pairs (S,p), where S is a subset of an n-person set and p belongs to S.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C4-R

How many shortest paths from (0,0) to (2,2) use only right and up moves?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C5-4

Choose any six distinct integers from 1,…,10. Prove that two chosen numbers sum to 11.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

C5-R

Nine objects go into four boxes. What occupancy must at least one box have?

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

N1-4

A number written in base 8 has digits $d_m \ldots d_0$. Derive a digit test for divisibility by 7 and one for divisibility by 9.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

N1-R

Is the decimal number 513 divisible by 9? Explain without long division.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

N2-4

Find all ordered pairs of positive integers a,b with gcd(a,b)=6 and a+b=54.

Working: _____

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Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

N2-R

Find gcd(48,18), showing the remainder steps.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

N3-4

Find the last digit of $7^{3^{2026}}$. Explain how you locate the exponent in the relevant cycle.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

N3-R

Find the last digit of 3^{12} and explain the zero-remainder case.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

N3-X1

Find the last digit of $7^{2026}+3^{2026}$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

N3-X2

Find the last digit of $7^{2026} \cdot 3^{2026}$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached

EX-VJ

Optional enrichment, with no baseline penalty: positive integers $a \le b$ satisfy $a^2+b^2+1=3ab$. For $a>1$, show that $b'=3a-b$ is a positive integer less than a and that (b',a) also solves the equation. Handle $a=1$ separately. Explain why repeated reduction reaches $(1,1)$.

Working: _____

_____

Confidence 0 / 1 / 2 / 3 · Familiarity: new / similar / exact · Status: attempted / not attempted / not reached